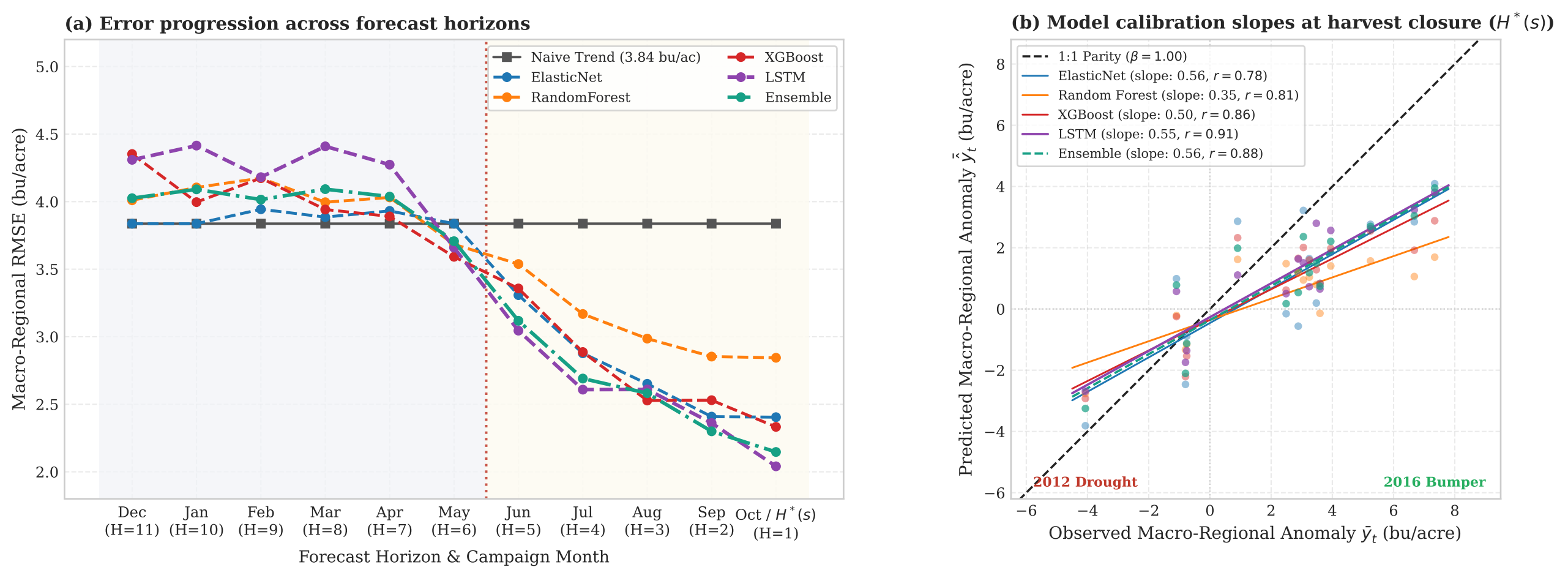
(a) Error progression across forecast horizons
Naive Trend (3.84 bu/ac)
ElasticNet
RandomForest
XGBoost
LSTM
Ensemble
Macro-Regional RMSE (bu/acre)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
Dec (H=11)
Jan (H=10)
Feb (H=9)
Mar (H=8)
Apr (H=7)
May (H=6)
Jun (H=5)
Jul (H=4)
Aug (H=3)
Sep (H=2)
Oct / $H^*(s)$ (H=1)
Forecast Horizon & Campaign Month
(b) Model calibration slopes at harvest closure ($H^*(s)$)
1:1 Parity ($\beta = 1.00$)
ElasticNet (slope: 0.56, $r = 0.78$)
Random Forest (slope: 0.35, $r = 0.81$)
XGBoost (slope: 0.50, $r = 0.86$)
LSTM (slope: 0.55, $r = 0.91$)
Ensemble (slope: 0.56, $r = 0.88$)
Predicted Macro-Regional Anomaly $\bar{\hat{y}}_t$ (bu/acre)
8
6
4
2
0
−2
−4
−6
2012 Drought
2016 Bumper
−6
−4
−2
0
2
4
6
8
Observed Macro-Regional Anomaly $\bar{y}_t$ (bu/acre)